Preface

Human Action Recognition (HAR) is a rapidly evolving field within computer vision and machine learning, driven by the increasing need for intelligent systems capable of understanding and interpreting human activities from video data. Applications of HAR span various domains, including surveillance, human-computer interaction, sports analytics, and healthcare, making it a critical area of research.

In this project, we present a novel approach to HAR by integrating two powerful models: Detectron2 and Long Short-Term Memory (LSTM) networks. Detectron2, a state-of-the-art object detection framework developed by Facebook AI Research, excels at identifying and localizing objects within images. Leveraging its robust capabilities, we extract high-quality, feature-rich representations of human actions from video frames. These features serve as the input to our second model, the LSTM network, which is well-suited for processing sequential data due to its ability to capture temporal dependencies.

The integration of Detectron2 and LSTM offers a comprehensive solution for HAR by combining spatial feature extraction with temporal sequence modeling. Our methodology involves first employing Detectron2 to detect and extract relevant features from each frame of a video. These features are then fed into the LSTM network, which processes the sequence of features to recognize and classify the human actions depicted in the video.

Through this project, we aim to demonstrate the effectiveness of combining advanced object detection and sequential modeling techniques for HAR. We evaluate our approach on benchmark datasets, highlighting its performance and potential applications in real-world scenarios. The results underscore the advantages of using Detectron2 for spatial feature extraction and LSTM for temporal analysis, paving the way for future research and development in human action recognition.